

POSTGRADUATE PROGRAMMES IN SYSTEMS ENGINEERING

University of Lagos Vision and Mission

Vision: To be a top class institution for the pursuit of excellence in knowledge, character and service to humanity.

Mission: To provide a conducive environment for teaching, learning, research and development, where staff and students will interact and compete effectively with their counterparts globally.

Systems Engineering Vision and Mission

Our mission and vision are same with that of the larger UNILAG institution. We produce versatile and multi-skilled engineers who are ever-ready and willing to contribute to societal development by optimally harnessing available resources.

Programme Structure

The MSc Systems Engineering programme consists of three key elements viz:

- A core component designed to provide the basic advanced knowledge and detailed technical skills required by the students to cover both numerical and symbolic based systems.
- An advanced taught component to provide further technical expertise in a chosen area of specialization.
- A project, designed to allow students to carry out a significant piece of work, either closely related to the requirements of the public or organised private sectors or at the leading edge of research.

To qualify for a degree a student must normally satisfy the examiners in all three components.

The taught programme comprises a minimum of four core courses and three elective courses chosen from a menu of several courses in the area of specialization of the student. Each course is equivalent to approximately 30 hours of lecture. **In** addition, each course will normally also involve group assignments and presentations.

The programme has five areas of specialization from which the student is expected to select one viz:

- Engineering Analysis
- Modelling and Simulation
- Artificial Intelligence

- Information and Communication
- Engineering Systems Management

SYSTEMS ENGINEERING PG SPECIALISATIONS

L DEGREE OF MASTER OF SCIENCE IN SYSTEMS ENGINEERING

Introduction

The M. Sc (Systems Engineering) is a programme consisting of courses of instruction and a project. Major areas of instruction are Continuum Mechanics, Engineering Modeling and Simulation, Computational Techniques, Stochastic Systems, Geophysical and Environmental Fluid mechanics. Others include software engineering, artificial intelligence Systems, Robotics, intelligent control and Water Resources, Applied Mechanics, Thermo fluids, Design and Production Engineering as well as Engineering Safety and Reliability.

These courses have been grouped into five designated areas of specialization as listed below:

- a) M. Sc (Systems Engineering) with specialization in Engineering Analysis
- b) M.Sc (Systems Engineering) with specialization in Modelling and Simulation
- c) M.Sc (Systems Engineering) with specialization in Artificial Intelligence
- d) M.Sc (Systems Engineering) with specialization in Information and communication Technology
- e) M.Sc (Systems Engineering) with specialization in Engineering Systems Management

For full-time students, the duration of the programme is twelve calendar months.

Admission Requirements:

(I). Candidates with a Bachelor's degree in an Engineering or Physical Sciences (Physics, Mathematics, Computer Science etc.) discipline from the University of Lagos or any other approved University may apply for admission into this programme.

(II). Candidates may be required to satisfy the Department in a selection process before admission.

(1II). Candidates are required to satisfy all other requirements of the School of Postgraduate Studies.

Degree Requirements

To satisfy the requirements for the award of the M.Sc. Systems Engineering degree, in any area of specialization, a candidate must: -

- (a). Offer and pass all core courses in his chosen area of specialization.
- (b). Offer and pass such number of 800-level elective courses as is required to bring the total of 800-level course units (including core courses but excluding project) to at least 21 units.
- (c). Offer and pass 6 units of project and

(d). Satisfy all other conditions stipulated in the regulation of the School of Postgraduate Studies.

D. PROGRAMME STRUCTURE

i, M.Sc. / M. Phil Courses

GENERAL CUMPULSORY COURSES

S/N	Course Code	Course Title	Units
1	SSG 801	Numerical Methods in Engineering	3
2	SSG 803	Methods of Applied Mathematics I	3
3	SSG 813	Linear and Non-Linear Systems Analysis	3
4	SSG 815	Knowledge-Based Systems and Artificial Intelligence	3

ii. M.Sc (Systems Engineering) with specialization in Engineering Analysis

a. Compulsory Courses

S/N	Course Code	Course Title	Units
1	SSG 804	Methods of Applied Mathematics II	3

b. Two elective courses from within the department, Faculty of Engineering or any other faculty within the University as may be appropriate for the designated area of specialization as listed below or as approved by the department.

S/N	Course Code	Course Title	Units
1	SSG 805	Methods of Continua	3
2	SSG 806	Mathematical Theory of Elasticity	3
3	SSG 807	Functions Space Methods	3
4	SSG 808	Continuum Mechanics and Thermodynamics of Solids	3

5	SSG 809	Electromagnetic Wave Interactions	3
6	SSG 810	Continuum Mechanics	3

iii. M.Sc. (Systems Engineering) with specialization in Modelling and Simulation

a. Compulsory Courses

S/N	Course Code	Course Title	Units
1	SSG 814	Systems Identification, Optimization and Control	3

b. Two Elective courses from within the department, Faculty of Engineering or any other faculty within the University as may be appropriate for the designated area of specialization as listed below or as approved by the department.

S/N	Course Code	Course Title	Units
1	SSG 811	Mathematical Programming	3
2	CSC 841	Simulation and Modelling	3

iv. M.Sc. (Systems Engineering) with specialization in Artificial Intelligence

a. Compulsory Courses

S/N	Course Code	Course Title	Units
1	SSG 816	Expert Systems with Applications to Engineering Problems	3
2	SSG 818	Introduction to Neural Networks and Applications	3

b. One elective course from within the department, Faculty of Engineering or any other faculty within the University as may be appropriate for the designated area of specialization as listed below or as approved by the department

S/N	Course Code	Course Title	Units
1	SSG 819	Robotic Applications	3
2	SSG 821	Expert Systems in Manufacturing and Engineering	3
3	SSG 831	Knowledge-Based Systems and Web Agents	3

v. M.Sc. (Systems Engineering) with specialization in Information and communication Technology

a. Compulsory Courses

S/N	Course Code	Course Title	Units
1	SSG 817	Communications and Networks	3
2	SSG 820	Fault-Tolerant Digital Systems	3

b. One elective course from within the department, Faculty of Engineering or any other faculty within the University as may be appropriate for the designated area of specialization as listed below or as approved by the department.

S/N	Course Code	Course Title	Units
1	SSG 831	Database Systems	3
2	CSC 811	Communication Networks and Distributed Processing	3
3	CSC 830	Data Base Management System Design	3

vi. M.Sc. (Systems Engineering with specialization in Engineering Systems Management

a. Compulsory Courses

S/N	Course Code	Course Title	Units
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1	SSG 845	Theory of Reliability of Engineering Systems	3
2	SSG 848	Crisis Management in Engineering Systems	3

b. One elective course from within the department, Faculty of Engineering or any other faculty within the University as may be appropriate for the designated area of specialization as listed below or as approved by the department.

S/N	Course Code	Course Title	Units
1	SSG 835	Hazardous Material Management I	3
2	SSG 837	Advanced Industrial Management	3
3	SSG 839	Occupational Safety Engineering II	3
4	SSG 837	Systems Methodology	3
5	SSG 841	Management Information Systems	3
6	SSG 842	Management and Methods in Reliability	3
7	SSG 843	Safety Engineering Management	3
8	SSG 844	Utility System Reliability	3
9	SSG 846	Knowledge Based Systems and Web Agents	3
10	SSG 847	Environmental Impact Assessment	3
11	SSG 849	Advanced Engineering Management Science	3

SSG 801: Numerical methods in Engineering (3 Units)

Methods for obtaining numerical solutions to problems arising in Engineering. Linear and non-linear mechanical systems. Ordinary and partial differential equations, initial-value problems, and extreme.; Calculus of variation. Function-space methods. Applications to vibrations, diffusion, heat transfer, wave propagation, membranes, plates, fluid flow and celestial mechanics. Simulation of dynamical system. Analog computation.

SSG 803: Methods of Applied Mathematics I (3 Units)

Ordinary differential equations; series; orthogonal functions and Sturm-Liouville theory; functions of several real variables, vector fields and integral theorems; matrices; partial differential equations.; Emphasis is on application and techniques of solution, wherever possible.

SSG 804: Methods of Applied Mathematics II (3 Units)

Continuation of partial differential equations; Green's function; Fourier and Laplace transforms; complex variables; calculus of variations; tensor analysis.

SSG 805: Mechanics of Continua (3 Units)

Introduction to the mechanics of continuous media. Cartesian tensors; kinematics of deformable media; stress, stress, balance laws; constitutive relations for selected solids and fluids. Further theory of kinematics of deformable media; thermodynamics and balance laws of continua; general theory of constitutive equations and typical examples; selected topics such as multipolar continuum theories, non-linear elasticity, variation principle for continuous fields, mixture theory, coupling of mechanical and electromagnetic fields.

SSG 806: Mathematical Theory of Elasticity (3 Units)

Development in tensor form of the basic equations of large-deformation elasticity; solution of certain large-deformation problems. Linearization to infinitesimal elasticity. Boussineq-Papkovich potentials and their application to three-dimensional problem; contact problems; plane stress by method of Muskhelishvili; application of conformal mapping; Cauchy integral techniques in elasticity; torsion problems.

SSG 807: Functions Space Methods (3 Units)

Function spaces in Engineering inner products.; Linear operators.; The Method of Moments; basic functions.; point matching.; Galerkin and Ritz procedures.; Approximate and Extended operators.; The numerical reduction of Engineering field problems.; System Matrices and Eigenfunction procedures.

SSG 808: Continuum Mechanics and Thermodynamics of Solids (3 Units)

Theory of nonlinear elasticity and Thermoelasticity; universal solutions, wave propagation, stability theory. Nonlinear viscoelasticity and introduction to more general theories of solids.

SSG 809: Electromagnetic Wave Interactions (3 Units)

Maxwell's Equations. Duality and Uniqueness. Image Theory, the Equivalence principle the Induction Theorem. Reciprocity. Green's Functions. Integral Equations. Plan, Cylindrical and Spherical waves. Radiation. Electromagnetic scattering. Aperture interactions. Perturbational and Network Permutations. Variational Techniques.

SSG 810: Continuum Electromagnetics (3 Units)

Static and quasi-stationary electromagnetic field equations. Field transformation in moving coordinate systems. Boundary conditions and constitutive relations for moving media. Magnetic diffusion and charge relaxation. the Maxwell Stress tensor. Electromagnetics of elastic solids and Newtonian fluids. Electromagnetic engineering hydrodynamics. Magnetic pumps, the linear conduction motor and magnetic levitation.

SSG 811: Communication Networks and Distributed Processing (3, units)

This course will cover major national and international network protocols, networked applications (applications of the Internet and Intranets), Data communications and networks (i.e., fundamentals of computer networks and networking technology), distributed systems (including the most important paradigms for building distributed applications including client-server computing, distributed object technology, and component models), security and management of networks and distributed systems (including fundamentals of computer and network security and investigates different approaches to network management).

SSG 813: Linear and Non-linear System Analysis (3 Units)

Numerical linear Algebra: Direct and Iterative Methods.; Generalised Eigenproblem.; Linear, multiple and quasi linear regression analysis.; Analysis of variance/covariance and design of orthogonal experiments. Unification of real differential systems, real difference systems and modular sequential systems. Application to critical path methods (CPM), PERT, network Analysis, production/inventory models and decision analysis. Introductory cybernetics and systems simulations. Digital filters, transfer functions, realizability, canonical forms, stability and minimal realisations.; properties of, and stability in, nonlinear network and systems.

SSG 814: System Identification, Optimization and Control (3 Units)

Numerical techniques for estimation/identification in static and dynamic models.; empirical dynamic models for linear sampled data systems with stochastic inputs. The Kalman Filter, its extensions and stochastic approximations. Classical and modal optimization methods. Mathematical programming search techniques: Linear, non linear and Dynamic models.

Optimal control for discrete and continuous time systems with equality and inequality constraints.

CSC 841: Modeling and Simulation (3 Units)

A study of the construction of models, which simulate real systems. The methodology of solution should include probability and distribution theory, statistical estimation and inference, the use of random varieties and validation procedures. A simulation language should be used for the solution of typical problems.

SSG 815: Knowledge-Based Systems & Artificial Intelligence (3 Units)

Basic concepts: Artificial intelligence; Intelligent machines and expert systems; symbolic representation of knowledge; symbolic processing with Prolog; Data dependency as an alternative paradigm to imperative programming; Rule-based problem solver and knowledge representation techniques implemented in Prolog; Examples based on use of grammars, production rule driven system and semantic nets. Machine intelligence.; Neural nets as an alternative to rule-based systems; The single-layer perceptron: the problem of linear separability; The multi-layer perceptron and the back-propagation algorithm; Heuristic Search and Inference; Forward and Backward chaining; Data Driven Reasoning; Goal-Driven Reasoning.; Heuristic Search: Domain-specific search; Min-max methods and strategies: pruning.

SSG 816: Expert Systems with Applications to Engineering Problems (3 Units)

Application of expert Systems to engineering problems using Prolog as the programming language. Engineering problems considered will be process control, systems and control. Applications will include electric utility systems, distillation columns and other process control problems.

SSG 818: Artificial Neural Networks

Artificial Neural Network (ANN) Architectures: backpropagation, variations on backpropagation networks, Kohonen self-organizing network, Adaline and Hopfield networks. This course also covers recurrent networks, stability considerations and performance optimization. Students experiment with the use of ANN in engineering through semester project.

SSG 819: Robotic Applications IT (3 Units)

History, development of the work environment for robots, their application and implementation. Concepts of control and sensory feedback in robots are covered.

SSG 821: Expert Systems in Manufacturing & Engineering (3 Units)

Intelligent engineering system design using knowledge bases, knowledge based problem solving, symbolic models, knowledge representation, inferencing are the topics covered. Students develop these skills through semester projects based on a specific engineering application using an expert system shell of their choice.

SSG 831: Knowledge Based Systems and Web Agents. (3 Units)

Introduction to knowledge based systems as a technique for problem solving in systems engineering. Knowledge representation, knowledge acquisition, search, logic and reasoning about uncertainty, analyses of the use of knowledge based systems for engineering design problems. Emphasis on the interface between operations research and artificial intelligence techniques for problem solving.

SSG 841: Data Base Systems (3 Units)

Fundamental concepts of database including a history of development, definition of terms, functional requirements of complex data structures, data base administrator functions, privacy-confidentiality issues, and future directions. Case studies are coordinated with a detailed examination of several commercially available systems.

SSG 817: Communication and Networks (3 units)

This course will cover fundamentals of digital communications and networking. Students will study the basics of information theory, sampling and quantization, coding, modulation, signal detection and system performance in the presence of noise. The study of data networking will include multiple access, reliable packet transmission, routing and physical protocols for data transmission and error detection/correction. The concepts taught in class will be discussed in the context of wireless communication.

SSG 820: Fault-Tolerant Digital Systems (3 Units)

Design and analysis of fault-tolerant digital systems. Fault models, hardware redundancy, information redundancy, evaluation techniques, system design procedures.

SSG 811: Communication Networks and Distributed Processing (3, units)

This course will cover major national and international network protocols, networked applications (applications of the Internet and Intranets), Data communications and networks (i.e., fundamentals of computer networks and networking technology), distributed systems (including the most important paradigms for building distributed applications including client-server computing, distributed object technology, and component models), security and management of networks and distributed systems (including

fundamentals of computer and network security and investigates different approaches to network management).

CSC 830: Database Management System Design (3 Units)

Database concepts: data independence; relationships, logical and physical organisations schema and subschema. Data models; hierarchical, network and relational. Data normalization. Data description languages. Query facilities. File organisation. Index organisation. File security. Data integrity and reliability.

SSG 835: Hazardous Material Management I (2 Units)

All aspects from generation to final disposal will be studied, including: identifying hazardous materials, chemical safety, storing and shipping chemicals, and treatment and disposal of hazardous wastes.

SSG 837 Advanced Industrial Management I (3 Units)

Contemporary management philosophies and their impact on engineers and engineering managers. topics include: total quality management, re-engineering, advanced quality initiatives, leadership and strategic planning.

SSG 839: Occupational Safety Engineering II (3 Units)

An overview of factors affecting safety in organizations, emphasizing analysis techniques and design strategies. Topics include occupational safety, accidents, fire protection, industrial hygiene, hazardous waste, toxicology, radiation safety, product liability and federal standards. A project involving a hazard analysis and the design of a solution for a field location is required.

SSG 840: Systems Methodology (3 Units)

This course introduces students to the basic processes for system design and engineering and examines the management of systems engineering projects. we first discuss the systems viewpoint and the systems engineering process. Next we examine issues relating to requirements specification and problem formulation. We highlight different methods for systems modelling and forecasting and for choosing among different system alternatives. The course also provides specific examples of systems engineering taken from the fields of telecommunications, manufacturing, environment! resource systems and transportation.

SSG 841 Management Information Systems (3 Units)

Study of the operational and managerial information needs of an organization. Emphasis is on the information needed throughout an organisation and on information systems to meet those needs.

SSG 842 Management and Methods in Reliability (3 Units)

Study of basic concepts in reliability as they apply to the efficient operation of industrial systems.

SSG 843 Safety Engineering Management (3 Units)

Principles of safety engineering applied to industrial situations; job safety analysis and specifications of solutions; reduction of accident rates, frequency and costs; protective equipment, jugs, fixtures and standards; rules, regulations and law.

SSG 844 Utility System Reliability (3 Units)

Reliability definition and measures. Probability concepts and Markov chains. Failure models and availability models. Utility system reliability. Loss of load probability method. Evaluation of transmission network reliability. Analysis of the electric power, water and telephone system reliability.

SSG 845 Theory of Reliability of Engineering Systems (3 Units)

Reliability Models; failure distributions, systems reliabilities, physical models, reliability design.

Analysis of failure data; empirical methods, reliability testing, identifying distributions. *Applications*; examples of analysis of failure/ repair data from the manufacturing, industrial and service sectors.

SSG 846 Knowledge Based Systems and Web Agents (3 Units)

Introduction to knowledge based systems as a technique for problem solving in systems engineering. Knowledge representation, knowledge acquisition, search, logic and reasoning about uncertainty, analyses of the use of knowledge based systems for engineering design problems. Emphasis on the interface between operations research and artificial intelligence techniques for problem solving.

SSG 847 Environmental Impact-Assessment (2 Units)

Introduction to environmental programs, the environmental impact assessment process, and technical, policy and legal implications. Emphasis will be placed on identification and evaluation of impacts, and attaining a sustainable future.

SSG 848 Crisis Management in Engineering Systems (3 Units)

Definition of Crisis (architecture and scale): 4-D Modelling of Crisis, Impact of Crisis on System: 1st and 2nd Order Effects: System Tools/features and role in crisis management; Monitoring and Surveillance, Forecast and Prediction Models; Contingency Plan; Disaster Mitigation and Relief Measures; Communication Networks; Decision Support Tools, Disaster Recovery Plans etc;

Uses of Case Studies: Analysis of recent cases of Oils Spills, Gas Pipeline Vandalism/Explosion, Dam Failure, Armoury Explosion etc.

SSG 849: Advanced Engineering Management Science (3 units)

Solving of managerial problems utilizing management science techniques. Problems are analysed, modelled and solved using such techniques as linear, goal, dynamic, programming, simulation, statistical analysis or other non-linear methods. Solutions will involve the use of computers. A study of the current literature in management science will also be conducted.

SSG 850: MSc Project (6 Units)

The student will be expected to carry out a significant piece of work either closely related to the requirements of the public or organised private sectors or at the leading edge of research.

Accademic, Technical and Administrative Staff-Department of Systems Engineering

S/N	NAME OF STAFF	QUALIFICATION	AREAS OF SPECIALIZATION
1	Dr. T. A Fashanu (Senior Lecturer)	B.Sc (Ife), M.Sc, Ph.D (Unilag), MAGU, MISEE, MASABE, MTMS,MNSE	Computational Visualisation Environmental Fluid Mechanics
2	Distinguished Emeritus Professor V.O.S. Olunloyo	B.Sc. Ph.D. (Cornell), FAEng, FAS, NNOM	Computational Fluid Mechanics
3	Distinguished Professor O. Ibidapo-Obe	B.Sc. (Unilag), M.Maths, Ph.D. (Waterloo), ASCE, FAAS, FAS, FAEng, OFR	StochasticProcesses,Artificial Intelligence, Simulation
4	Professor O. A. Fakinlede	B.Sc. (Unilag), Ph.D. (Alberta), R.Engr.	Computational Mechanics, Software Engineering
5	Professor O. Taiwo	B.Sc (UCL),PhD (UMIST)	Process Systems Engineering
6	Dr. O.S. Asaolu (Senior Lecturer)	B.Sc, M.Sc, Ph.D (Unilag), FIOR, MGYA, MNYA, MNCS, MIIE, MNSE, R.Engr.	Artificial Intelligence, Software Engineering, Systems Modelling& Analysis
7	Dr. F.O. Ogunwolu (Senior Lecturer)	B.Sc, M.Sc, Ph.D (Unilag), FIOR	Modelling& Simulation, Optimization Techniques, Discrete Systems Engineering.

8	Dr. C. A. Osheku (Senior Lecturer)	B.Sc. (Ife), M.Sc., Ph.D. (Unilag), MNSE, R.Engr.	Computational Fluid Mechanics/Aerodynamics
9	Dr. O.O.E. Ajibola (Senior Lecturer)	B.Sc (UI), M.Sc, Ph.D (Unilag), MMAN, MNSA, MSESN	Artificial Intelligence (Biomedical Modelling)
10	Dr. A.M. Ajofoyinbo (Senior Lecturer)	B.Sc. (UI), M.Sc., M.Sc., Ph.D (Unilag), AMNSE, CPRCN, MIEEE, MNES	Intelligent control & automation; Telecommunication Systems
11	Dr. M.K.O Ayomoh (Senior Lecturer)	B.Eng(Ilorin), M.Sc (UI), Ph.D(Unilag), MNSE, R.Engr.	Artificial Intelligence (Robotics)
S/N	NAME OF STAFF	QUALIFICATION	AREAS OF SPECIALIZATION
12	Dr. O.A. Alli (Lecturer I)	B.Eng (FUTA), M.Sc (UI), PhD (Unilag) MNSE, MNGS, R.Engr.	Industrial /Production Engineering, Excavation Dynamics
13	Dr. O.P. Popoola (Lecturer I)	B.Sc., M.Sc (UI), Ph.D(China)	Pattern Recognition and Intelligent Systems
14	Engr. T. T. Akano (Lecturer II)	B.Eng (FUTO), M.Sc (Unilag), MNSE, MNIMEch, R.Engr.	Computational Mechanics, Software Engineering
15	Mr. O.A. Adeleye (Lecturer II)	B.Eng (FUTA), M.Sc (Unilag), ASME, R.Engr.	Computational Mechanics & Engineering Analysis
16	Engr .A. A. Sosimi (Lecturer II)	B.Sc., M.Sc., (UNILAG) MNSE, R.Engr.	Artificial Intelligence (Computational Linguistics& Natural Language Processing)
17	Engr. O.Ipinnimo (Lecturer II)	B.Eng (UNAD), M.Sc (Unilag), MNSE, R.Engr.	Electrical/ Electronics
18	Engr. J. Ogbemhe (Lecturer II)	B.Eng (FUTA), M.Sc. (Unilag), MNSE, R.Engr.	Artificial Intelligence (Robotics)
19	Mr. C. Onyedikam (Asst. Lecturer)	B.Sc., M.Sc. (Unilag), AMIOR	Transportation Modelling and Planning, Optimization

			techniques, Systems Modelling & Simulation
20	Mr. O.I. Oluwasuji (Asst. Lecturer)	B.Sc,(LAUTECH), M.Sc. (Newcastle, U.K)	Automation and Control
21	Mr. K. O. Orolu (Asst. Lecturer)	B.Sc, M.Sc,(UNILAG)	Intelligent control

Technical Staff

S/N	NAME OF STAFF	Designation	Qualification
1	Ms Onuoha, Blessing Ogechi	Research Technologist	B.Sc., Systems Engineering
2	Mr Olatunji, Musibau Jamiu	Research Technologist	B.Sc., Systems Engineering
3	Mr Olowosulu, Emmanuel Oluwaseyi	System Administrator	B.Sc., Systems Engineering

Administrative staff

S/N	Name of Staff	Designation	Qualification
1	Mrs Fatima Moriselade Ayoade	Confidential Secretary	HND in Secretarial Administration
2	Mrs. Felicia Iyabo Ayeni	Senior Data Processing Officer	Advanced Diploma in Computer Studies
3	Mrs. Sade Sanusi	Chief Clerical Officer	Diploma in Corporate Admin. & Registry Mgt
4	Mr. Oloruntoba Femi Oladipupo	Driver	G.C.E/O Level